Royal Devon University Healthcare NHS Foundation Trust

Guideline for the use of:
Activated Charcoal Filters on anaesthetic machines

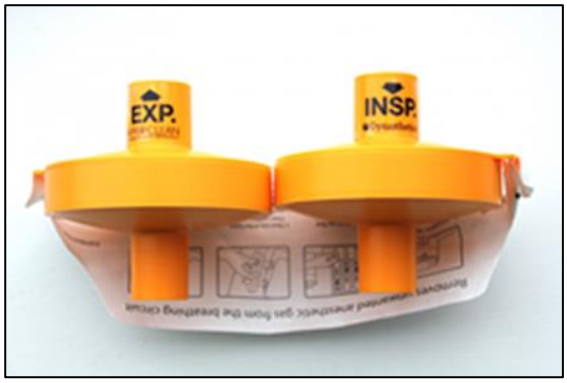
Summary

This guideline is for all theatre staff involved in the management of patients with Malignant Hyperthermia (MH) and details the appropriate use of activated charcoal filters on anaesthetic machines in two scenarios:

1. Emergency management of MH crisis under general anaesthesia
2. Preparation of the anaesthetic machine prior to anaesthetising a patient with known or suspected MH susceptibility

Key Points

Malignant hyperthermia (MH) crisis management recommendations advocate the use of activated charcoal filters. This guideline details the appropriate use of the filters in emergency MH management and in the routine preparation of the anaesthetic machine for MH susceptible patients.



PUBLICATION DETAILS

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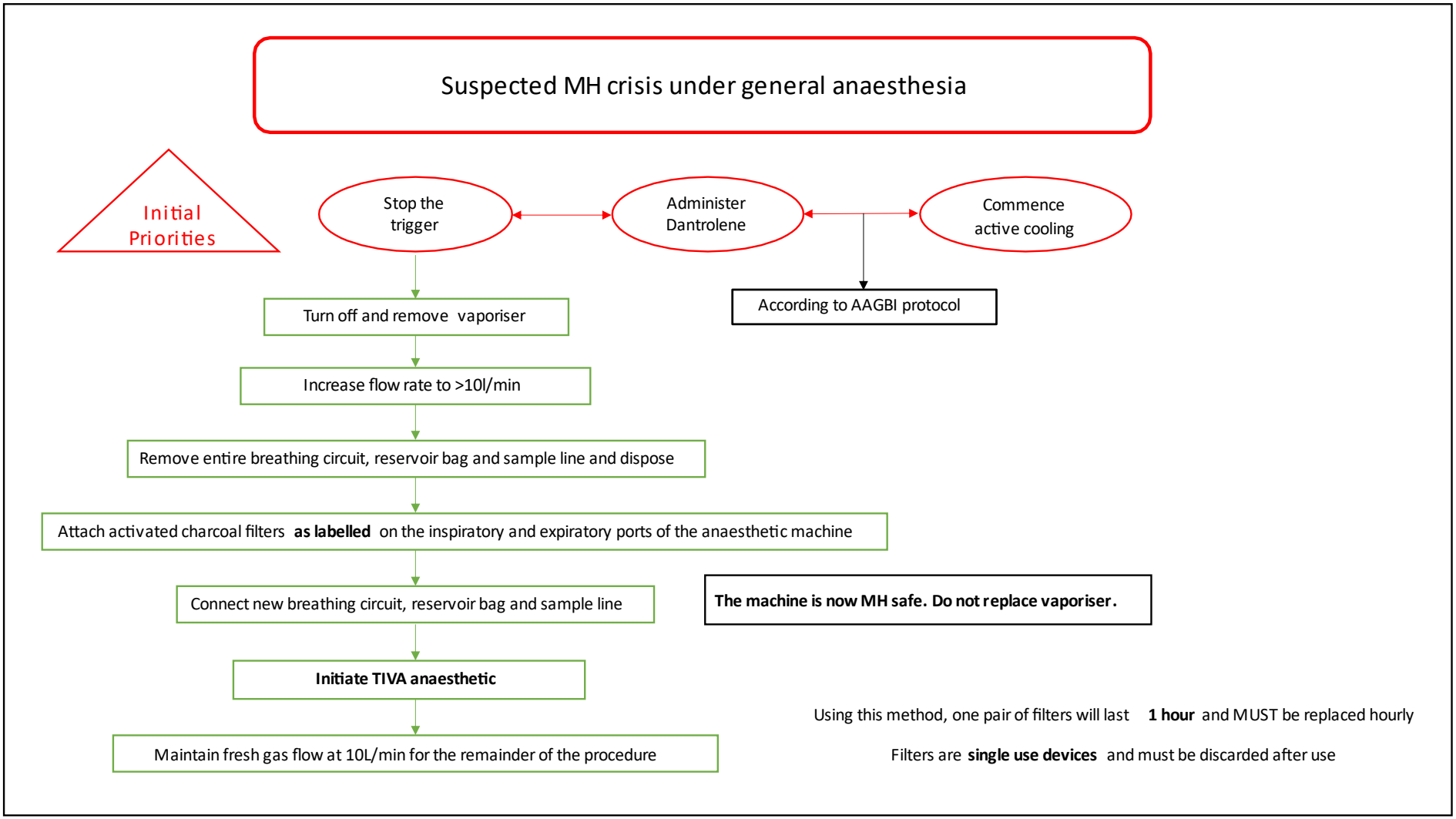
Associated Trust Policies and References:

- Emergency Action Checklists ESA emergency anaesthesia checklists: modified for UK use in Exeter
- Quick Reference Handbook AAGBI Guidelines for crises in anaesthesia
- Hopkins, P.M., Girard, T., Dalay, S., Jenkins, B., Thacker, A., Patteril, M. and McGrady, E. (2021), Malignant hyperthermia 2020. Anaesthesia. https://doi.org/10.1111/anae.15317

FOR PATIENTS WITH SUSPECTED MH CRISIS UNDER GENERAL ANAESTHESIA



FOR PATIENTS WITH KNOWN OR SUSPECTED MH SUSCEPTIBILITY

Machine preparation for patients with known or suspected MH susceptibility

1. Remove entire breathing circuit, reservoir bag and sample line and dispose
2. Remove vaporisers
3. Flush anaesthetic machine with >10l/min oxygen for 90 secs
4. Attach activated charcoal filters **as labelled** on the inspiratory and expiratory ports of the anaesthetic machine
5. Connect new breathing circuit, reservoir bag and sample line
6. **Initiate TIVA anaesthetic**
7. Maintain fresh gas flow at 3L/min for the remainder of the procedure

The machine is now MH safe.

Do not replace vaporiser.

Using this method, one pair of filters will last **12 hours** and MUST be replaced after this time

Filters are **single use devices** and must be discarded after use

1. INTRODUCTION

1.1 Malignant Hyperthermia (MH) is a rare complication of general anaesthesia, with an estimated mortality rate of 4%. It presents as a hypermetabolic response to triggering agents – suxamethonium and volatile anaesthetic agents – in genetically susceptible individuals.

1.2 The comprehensive management of MH crisis includes:

- Elimination of the triggering agent
- Administration of Dantrolene
- Active body cooling.

These strategies should be carried out by the multi-disciplinary theatre team, in accordance with established emergency protocols and prior training.

1.3 The use of activated charcoal filters forms part of the process to eliminate the triggering volatile agent from the anaesthetic machine.

2. PURPOSE

2.1 This document details the appropriate use of activated charcoal filters in two scenarios:

- Emergency management of MH crisis under general anaesthesia.
- Preparation of the anaesthetic machine prior to anaesthetising a patient with known or suspected MH susceptibility.

3. DEFINITIONS

3.1 **Malignant Hyperthermia** – hypermetabolic crisis in response to triggering anaesthetic agents in genetically susceptible individuals.

4. DUTIES AND RESPONSIBILITIES OF STAFF

4.1 The **anaesthetist** is responsible for identifying MH susceptible patients during the pre-operative assessment.

4.2 The **ODP** is responsible for stocking the filters and ensuring they are in date.

4.3 The **anaesthetist or ODP** is responsible for attaching the filters to the machine and replacing them at the appropriate time intervals.

4.4 The **ODP** is responsible for disposal of the filters after use.

5. LOCAL OPERATING PROCEDURE FOR ACTIVATED CHARCOAL FILTERS

5.1 **Emergency use** of activated charcoal filters for **suspected MH crisis** during general anaesthesia:

1. Turn off vaporiser (remove if desired)
2. Do not delay administration of Dantrolene
3. Increase flow rate to >10L/min
4. Remove entire breathing circuit, reservoir bag and sample line and dispose

5. Attach activated charcoal filters **as labelled** on the inspiratory and expiratory ports of the anaesthetic machine
6. Connect new breathing circuit, reservoir bag and sample line
7. The machine is now MH safe. Do not replace vaporiser or switch on
8. **Initiate TIVA anaesthetic**
9. Maintain fresh gas flow at 10L/min for the remainder of the procedure

Using this method, a pair of filters will last **1 hour** and **MUST** be replaced hourly

Filters are **single use devices** and must be discarded after use.

5.2 **Routine use** of activated charcoal filters for patients with known or suspected MH susceptibility:

1. Remove entire breathing circuit, reservoir bag and sample line and dispose
2. Remove vaporisers (optional)
3. Flush anaesthetic machine with **>10L/min oxygen** for **90 seconds**
4. Attach activated charcoal filters **as labelled** on the inspiratory and expiratory ports of the anaesthetic machine
5. Connect new breathing circuit, reservoir bag and sample line
6. The machine is now MH safe. Do not replace vaporiser or switch on
7. **Initiate TIVA anaesthetic**
8. Maintain fresh gas flow at 3L/min throughout case

Using this method, a pair of filters will last **12 hours** and **MUST** be replaced after this time

Filters are **single use devices** and must be discarded after use.

6. STAFF TRAINING

6.1 Actions will be undertaken to ensure all relevant staff are aware of this guideline and the appropriate use of activated charcoal filters for the proactive or reactive anaesthetic management of malignant hyperthermia. An initial email cascade was circulated to relevant staff. For new staff going forward, training will take place as part of local induction processes.

6.2 Training by a company representative covering the following specific issues:

- Selecting cases for using activated charcoal filters.
- Appropriate application of filters to anaesthetic machine.
- Duration of action for each set of filters.

6.3 Flowsheet attached to anaesthetic machines.
Information kept with MH kit and emergency guidelines.

6.4 Training videos available.